Example:

Glutamate + Oxaloacetate $\xrightarrow{\text{AST}}$ α-Ketoglutarate + Aspartate

- It is involved in the formation of heme in hemoglobin.
- Pyridoxal phosphate acts as a coenzyme for the decarboxylase in decarboxylation reactions.Amino acids are decarboxylated to form their corresponding amines.

Histidine $\xrightarrow[B_6 \text{ - phosphate}]{\text{Decarboxylase}}$ Histamine

- Vitamin B_6 is involved in the synthesis of coenzyme A from pantothenic acid.
- It is also involved in the production of antibodies.
- Pyridoxal phosphate plays a key role in glycogenolysis.

Absorption and storage

Pyridoxine is readily absorbed by the small intestine. The excess amount if ingested is not stored in the body, and is excreted in urine.

Deficiency:

Deficiency of vitamin B6 is extremely rare. In infants its deficiency causes irritabilities, insomnia, muscular weakness and convulsion. The cause of the convulsions are severe impairment of the activity of the enzyme glutanate ecarboxylase, which is dependant on pyridoxal phosphate. The product of Glutamate decarboxcylase in γ-aminobutyric acid (GABA), which is a reguletory neurotransmitter in the central nervous system.

v. Vitamin B_{12}:

Vitamin B_{12} has complex ring structure, similar to porphyrin ring which has a cobalt ion (Co^{3+}) at its center (Figure. 8.23). The active coenzyme forms of vitamin B_{12} are methylcobalamine and deoxyadenolsyl cobalamine. The six co-ordination valencies of cobalt ion (Co^{3+}) are satisfied by the four nitrogens of reduced terapyrrole and the fifth by the nitrogen of 5, 6 dimethyl benzimidazole moiety and the sixth one either by CN^- (cyanocobalamine) or H_2O (aquacobalamine) or OH^- (Hydroxycobalamine) or CH_3 (methylcobalamine)

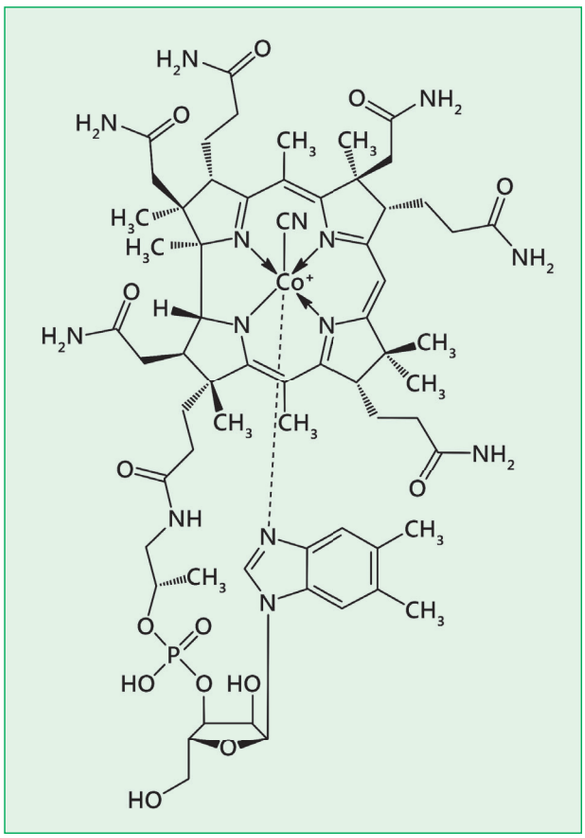


Figure 8.23 Structure of Vitamin B_{12}

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